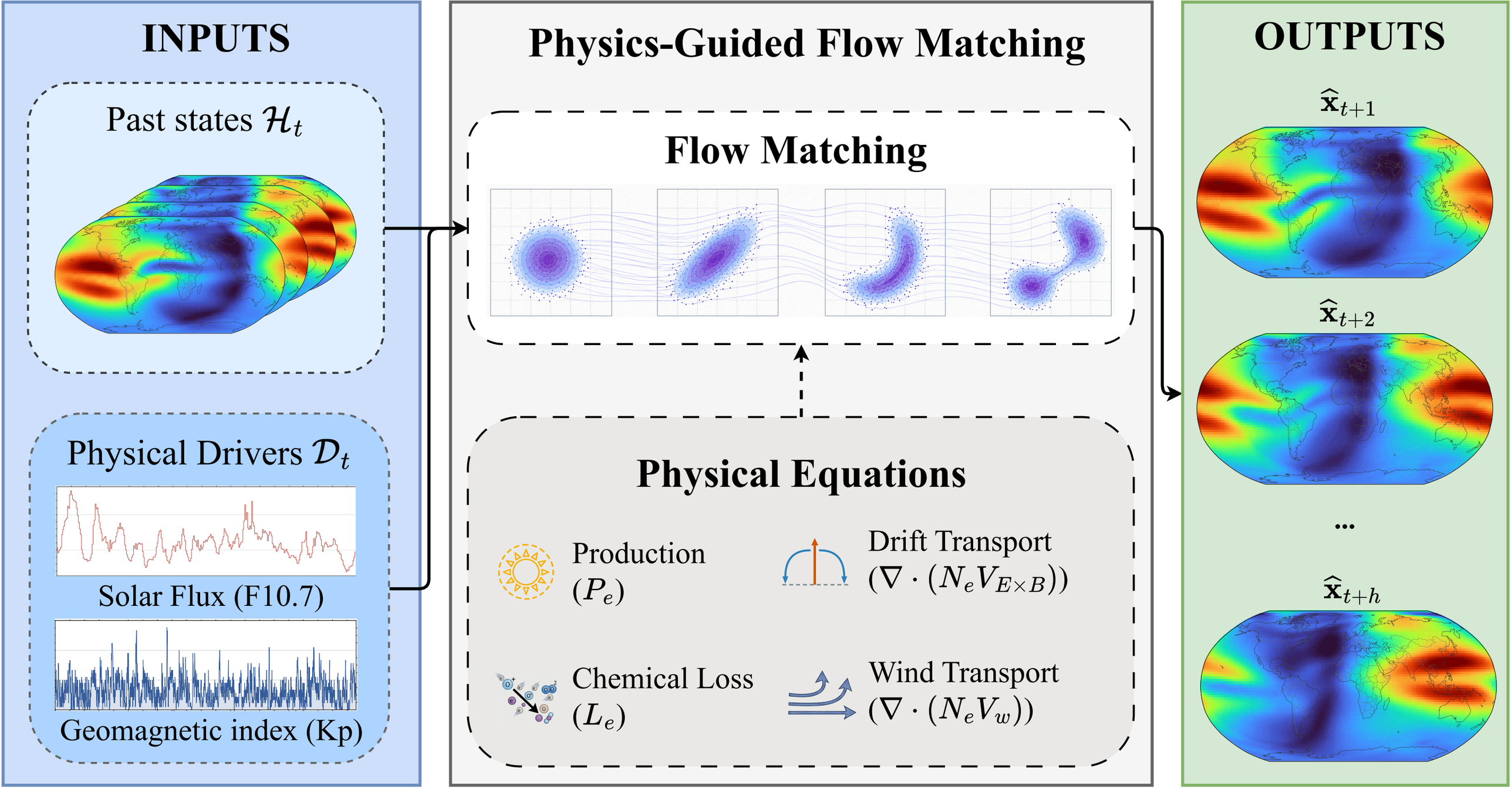

INPUTS
Past states $\mathcal{H}_t$
Physical Drivers $\mathcal{D}_t$
Solar Flux (F10.7)
Geomagnetic index (Kp)
Physics-Guided Flow Matching
Flow Matching
Physical Equations
Production (P_e)
Drift Transport ($\nabla \cdot (N_e V_{E\times B})$)
Chemical Loss (L_e)
Wind Transport ($\nabla \cdot (N_e V_w)$)
OUTPUTS
$\hat{\mathbf{x}}_{t+1}$
$\hat{\mathbf{x}}_{t+2}$
...
$\hat{\mathbf{x}}_{t+h}$